Richard Teague

Kerr–McGee Development Assistant Professor
Massachusetts Institute of Technology

rteague@mit.edu
pfl.mit.edu
(+1) 617-495-7259
Cambridge, Massachusetts

EDUCATION

2013–2017 **Ph.D. in Astronomy (Magna Cum Laude) — Max-Planck-Institute for Astronomy**
Heidelberg, Germany

2008–2013 **MPhys Astrophysics (First Class Honours) — University of Edinburgh**
Edinburgh, United Kingdom

ACADEMIC APPOINTMENTS

07/22–present **Kerr–McGee Development Assistant Professor**
Massachusetts Institute of Technology · Dept. of Earth, Atmospheric and Planetary Sciences

05/22–04/25 **Research Associate**
Smithsonian Astrophysical Observatory

09/19–04/22 **Submillimeter Array Fellow**
Center for Astrophysics · Harvard & Smithsonian

05/17–07/19 **Postdoctoral Researcher**
University of Michigan

01/17–04/17 **Postdoctoral Researcher**
Max-Planck-Institute for Astronomy

AWARDS & HONORS

2026 **AAS Fred Kavli Plenary Speaker** — for pioneering work revealing the 3D kinematic velocity fields within planet-forming disks

2024 **EAPS Community Builder Award** — recognition of an EAPS community member for exceptional service to EAPS

09/22 **pH Lectureship** — recognizes a CfA scientist who shows exceptional promise early in their career

03/20 **Harvard Data Science Initiative Research Fund** — Regularized Maximum Likelihood Imaging: A New Method for Detecting Planets

11/16 **Ernst Patzer Award** — awarded for the best refereed publication by a young scientist

2010, 2011 — **Pre-Honours Certificate of Merit** — top 5% performance in pre-honours exams, University of Edinburgh

OBSERVATIONAL TIME & TELESCOPE PROPOSALS (AS PI)

Over 392 hours (600+ as co-I) awarded on ALMA as PI, including the exoALMA Large Program; 20 hours (165 as co-I) on IRAM; 46 hours (30 as co-I) on the SMA; 26 hours (18 as co-I) on JWST as co-PI; 6 hours on VLT/GRAVITY; 3 nights (9 as co-I) on Magellan. Selected PI proposals below.

2021 — **exoALMA Large Program**
ALMA, 183 hours, 2021.1.01123.L, A ranked · PI (co-PIs: Benisty, Facchini, Fukagawa & Pinte)

2025 — **Detecting and Characterizing Magnetic Fields in Two Disks through Zeeman Splitting**
ALMA, 47 hours, 2025.1.00304.S, B ranked · PI

2025 — **Calibrating Disk Mass Measurement Techniques**
ALMA, 14 hours, 2025.1.00940.S, A ranked · PI

2022 — **The Most Sensitive Search for Magnetic Fields in a Solar Nebula Analogue**
ALMA, 18 hours, 2022.1.00840.S, A ranked · PI

2021 — **Is the Magneto-Rotational Instability Driving Protoplanetary Disk Evolution?**
SMA, 30 hours, 2020A-S033, A ranked · PI

2019 — **Mapping the 3D Kinematic Structure of Planet Formation**
ALMA, 33.2 hours, 2019.1.00419.S, B ranked · PI

2018 — **An Unambiguous Detection of a Magnetic Field in a Protoplanetary Disk**
ALMA, 6.7 hours, 2018.1.00980.S, A ranked · PI

INVITED TALKS & COLLOQUIA (SELECTED)

60+ invited talks and colloquia since 2015; a selection of recent talks is listed below

03/26 — **“From Pebbles to Planets: New Frontiers in Planet Formation”** — Caltech GPS Division Seminar

02/26 — **“Searching for Turbulence: Methods, Limitations and Potential”** — Flatiron Workshop on Hydrodynamic and Dusty Turbulence in Protoplanetary Disks

01/26 — **“The Search for Magnetic Fields in Protoplanetary Disks”** — MPIA & ESO Star and Planet Formation Seminars

05/25 — **“An Overview of the exoALMA Program”** — COST Action: PLANETS Seminar

04/25–03/25 — **“How To Build a Solar System”** — Johns Hopkins & STScI Joint Colloquium; UC Berkeley Astronomy Colloquium

06/24 — **“An Interferometric View of the Planet Formation Environment: Past, Present and Future”** — GRAVITY+: Impact on Star and Planet Formation

03/24–02/24	**“From Pebbles to Planets: What Gas Can Tell Us”** — McMaster, CITA & Munich Joint Astronomical Colloquia
12/23	**“The Dynamical Structure of Planet Forming Disks”** — ALMA at 10 Years: Past, Present, and Future
02/23–06/23	**“Witnessing the Formation of Giant Planets and their Moons”** — Gordon Conference on the Origins of Solar Systems; Boston University; MIT Haystack; Ohio State; Harvard EPS Colloquia
09/22	**“Exploring the Youngest Planetary Systems”** — Center for Astrophysics | Harvard & Smithsonian pH Lecture
2019–2021	**“Witnessing the Assembly of Planetary Systems” / “Mapping the Assembly of Planetary Systems in 6 Dimensions”** — CfA, Munich, ETH Zurich, Cambridge, McMaster, Caltech & JPL Colloquia

TEACHING

2022–present	**12.410 — Observational Techniques for Optical Astronomy** Instructor
2025–present	**12.423 — Planet Formation** Instructor
2024	**12.S681 — From Grains to Gas Giants: The Formation of Planetary Systems** Instructor
2014	**Wavefront Analysis Laboratory** Instructor

ADVISING & MENTORING

Ph.D. students (MIT)	**Leah Albrow** (2024–present) · **Isabella Macias** (2024–present) · **Jensen Pierce Thomas Lawrence** (2023–present)
Masters students	**Marco Duraku, Imperial College London** (2024–present, MIT–Imperial Exchange) · **Amius Marshall De’Ath, Imperial College London** (2024–2025) · **Abigail Colclasure, MIT** (2024–2025)
Undergraduates (MIT)	**Jackson Holland** (2026–present) · **Erin Cusson** (2024–present) · **Anika Nath** (2024) · **Carol Chen** (2022–2023) · **Aidan van Duzer** (2023) · **Anna Orgel** (2022–2023)
Academic advisor	**Jan Toomlaid** (2024–present) · **Kaylee Barrera** (2023–present) · **Anika Nath** (2023–2025)
Prior institutions	Co-supervised graduate and undergraduate students at the University of Michigan (Felipe Alcaron, Jenny Calahan, Deryl Long, Case Hazewinkel, Jeanne Kwon, 2018–2020), Harvard University (Alessandra Canta, 2020–2021), Beijing Normal University (Haochuan Yu, 2020–2022) and Ludwig Maximilian University (Julian Penzinger, 2016, 2018)

PROFESSIONAL SERVICE

Refereeing	**Referee for AAS journals, *A&A*, *MNRAS* and *Nature* journals** · **NESSF external reviewer** (2018, 2020)
Departmental (MIT/EAPS)	**EAPS Council, Junior Faculty Representative** (2025–present) · **CORE: Community, Outreach and Respect in EAPS** (2025–present) · **EAPS Department Lecture Series Planetary Science Representative** (2025–present) · **EAPS Committee on the Education Program** (2023–2024) · **EAPS Distinguished Postdoctoral Fellowship Committee** (2022)
Workshop & conference SOCs	**Astrochemistry in the Broadband Era: ngVLA and ALMA WSU** (2025) · **Spatio-spectral Modeling of ALMA Data Cubes: ALMA-2030** (2024) · **exoALMA Start of Science Workshop** (2022) · **Vertical Shear Instability Meeting** (2022) · **SMA Interferometry School** (2021) · **Visualizing the Kinematics of Planet Formation, Flatiron Institute** (2019)
Diversity, equity & inclusion	**Postdoc and Research Scientist DEI Representative** (2018–2019) · **Conversations on Equity and Inclusion Co-organizer** (2018–2019) · **Equi-Tea Organizer** (2018–2019)
Seminar organizing	**SMA Seminar Organizer** (2020–2021) · **Stars, Planets and Formation Seminar Organizer, UMich** (2018–2019) · **MPIA Student Representative & Workshop Organizer** (2015–2017) · **IMPRS Graduate Student Representative** (2013–2017)
Outreach	**“How to Find Baby Planets,” University of Michigan Lowbrow Astronomers** (2020)

REFEREED PUBLICATIONS

179 refereed articles · 24 as first author · reverse chronological · name in bold

2026

20 PAPERS

179 **First Detection of HC_5N in a Class II Disk around TW Hya**
Wampler, S. C., R. A. Loomis, A. Diop, et al.
The Astrophysical Journal, 1003, L30

178 **The circumbinary disc of HD 34700A: II. Analysis of a strong dust asymmetry**
Fasano, D., M. Benisty, J. Stadler, et al.
Astronomy and Astrophysics, 709, A174

177 **An SMA Molecular Inventory of the Edge-on Protoplanetary Disk Gomez's Hamburger**
Cusson, E. M., L. Wölfer, **R. Teague**, et al.
arXiv e-prints, arXiv:2605.26512

176 **Probing Dust in the MWC 480 Disk from Millimeter to Centimeter Wavelengths**
Shi, Y., F. Long, E. Macías, et al.
The Astrophysical Journal, 1001, 27

175 **exoALMA. XXI. The Morphology and Dynamics of Vertical Flows**
Benisty, M., A. F. Izquierdo, J. Stadler, et al.
The Astrophysical Journal, 1000, L14

174 **exoALMA. XXIII. Estimating Disk and Planet Properties from Dust Morphologies with DBNets 2.0**
Ruzza, A., G. Lodato, G. Rosotti, et al.
The Astrophysical Journal, 1000, L16

173 **exoALMA. XX. Tomographic Detection of Embedded Planets in Protoplanetary Disks**
Izquierdo, A. F., J. Bae, S. Facchini, et al.
The Astrophysical Journal, 1000, L13

172 **exoALMA XXII: A Two-dimensional Atlas of Deviations from Keplerian Disks**
Fukagawa, M., A. F. Izquierdo, J. Stadler, et al.
The Astrophysical Journal, 1000, L15

171 **exoALMA. XXIV. Formaldehyde Emission in Protoplanetary Disks of exoALMA Compared with Their Properties and Dynamical State**
Alarcón, F., S. Facchini, L. Trapman, et al.
The Astrophysical Journal, 1000, L32

170 **AB Aur, a Rosetta stone for studies of planet formation: IV. C/O estimates from CS and SO interferometric observations**
Rivière-Marichalar, P., A. Fuente, R. le Gal, et al.
Astronomy and Astrophysics, 707, A348

169 **Benchmarking pre-main sequence stellar evolutionary tracks using disk-based dynamical stellar masses**
Zallio, L., M. Vioque, S. M. Andrews, et al.
Astronomy and Astrophysics, 708, L1

168 **The circumbinary disk of HD34700A: I. CO gas kinematics indicate spirals, infall, and vortex motions**
Stadler, J., M. Benisty, F. Zagaria, et al.
Astronomy and Astrophysics, 707, A160

167 **A Protoplanet Candidate in the PDS 66 Disk Indicated by Silicon Sulfide Isotopologues**
Yoshida, T. C., F. Alarcón, J. Bae, et al.
The Astrophysical Journal, 999, L22

166 **A 2 au Resolution View by ALMA of the Planet-hosting WISPIT 2 Disk**
Facchini, S., P. Curone, M. Benisty, et al.
The Astrophysical Journal, 998, L16

165 **exoALMA. XIX. Confirmation of Non-thermal Line Broadening in the DM Tau Protoplanetary Disk**
Hardiman, C., C. Pinte, D. J. Price, et al.
The Astrophysical Journal, 997, L47

164 **Tracing Pebble Drift History in Two Protoplanetary Disks with CO Enhancement**
Armitage, T., J. Williams, K. Zhang, et al.
The Astrophysical Journal, 998, 308

163 **A Submillimeter Survey of CS Excitation in Protoplanetary Disks: Evidence of X-Ray-driven Sulfur Chemistry**
Law, C. J., R. Le Gal, K. I. Öberg, et al.
The Astrophysical Journal, 997, 91

162 **Astrometric view of companions in the inner dust cavities of protoplanetary discs**
Vioque, M., R. A. Booth, E. Ragusa, et al.
Astronomy and Astrophysics, 705, A238

161 **The ^{12}CO gas structures of protoplanetary disks in the Upper Scorpius region**
Zallio, L., G. P. Rosotti, M. Vioque, et al.
Astronomy and Astrophysics, 705, A49

160 **Physical and Chemical Characterization of GY 91's Multi-ringed Protostellar Disk with ALMA**
Jiang, S. D., J. Huang, I. Czekala, et al.
arXiv e-prints, arXiv:2601.18884

2025

31 PAPERS

159 **Direct Measurement of Extinction in a Planet-hosting Gap**
Cugno, G., S. Facchini, F. Alarcon, et al.
The Astronomical Journal, 170, 317

158 **Satellites and Small Bodies With ALMA: Insights Into Solar System Formation and Evolution**
de Kleer, K., M. E. Brown, M. Cordiner, et al.
AGU Advances, 6, e2025AV001778

157 **Spirals and Vertical Motions in the Planet-Forming Disk around HD 100546. A multi-line study of its gas kinematics**
Wölfer, L., A. F. Izquierdo, A. Booth, et al.
arXiv e-prints, arXiv:2512.13814

156 **Winding motion of spirals in a gravitationally unstable protoplanetary disk**
Yoshida, T. C., H. Nomura, K. Doi, et al.
Nature Astronomy, 9, 1672

155 **JWST-MIRI Observations of the Irradiated Chemistry in the Inner Disk Cavity of GM Aur**
Romero-Mirza, C. E., K. I. Öberg, A. Banzatti, et al.
The Astrophysical Journal, 991, 128

154 **The Light Echo of a High-redshift Quasar Mapped with Lyα Tomography**
Eilers, A.-C., M. Yue, J. Matthee, et al.
The Astrophysical Journal, 991, L40

153 **Revealing Fine Structure in Protoplanetary Disks with Physics Constrained Neural Fields**
Levis, A., N. Luong, **R. Teague**, et al.
arXiv e-prints, arXiv:2509.03623

152 **A Radially Resolved Magnetic Field Threading the Disk of TW Hya**
Teague, R., B. Lankhaar, S. M. Andrews, et al.
The Astrophysical Journal, 991, L6

151 **exoALMA. XVIII. Interpreting Large-scale Kinematic Structures as Moderate Warping**
Winter, A. J., M. Benisty, A. F. Izquierdo, et al.
The Astrophysical Journal, 990, L10

150 **Leaky dust trap in the PDS 70 disc revealed by ALMA Band 9 observations**
Sierra, A., M. Benisty, P. Pinilla, et al.
Monthly Notices of the Royal Astronomical Society, 541, 3101

149 **SO Emission in the Dynamically Perturbed Protoplanetary Disks around CQ Tau and MWC 758**
Zagaria, F., H. Jiang, G. Cataldi, et al.
The Astrophysical Journal, 989, 30

148 **Inner disc and circumplanetary material in the PDS 70 system: Insights from multi-epoch, multi-frequency ALMA observations**
Fasano, D., M. Benisty, P. Curone, et al.
Astronomy and Astrophysics, 699, A373

147 **Radial variations in the nitrogen, carbon, and hydrogen fractionation in the PDS 70 planet-hosting disk**
Rampinelli, L., S. Facchini, M. Leemker, et al.
Astronomy and Astrophysics, 698, A115

146 **exoALMA. XIV. Gas Surface Densities in the RX J1604.3−2130 A Disk from Pressure-broadened CO Line Wings**
Yoshida, T. C., P. Curone, J. Stadler, et al.
The Astrophysical Journal, 984, L19

145 **exoALMA. XV. Interpreting the Height of CO Emission Layer**
Rosotti, G. P., C. Longarini, T. Paneque-Carreño, et al.
The Astrophysical Journal, 984, L20

144 **exoALMA. XVI. Predicting Signatures of Large-scale Turbulence in Protoplanetary Disks**
Barraza-Alfaro, M., M. Flock, W. Béthune, et al.
The Astrophysical Journal, 984, L21

143 **exoALMA. XVII. Characterizing the Gas Dynamics around Dust Asymmetries**
Wölfer, L., M. Barraza-Alfaro, **R. Teague**, et al.
The Astrophysical Journal, 984, L22

142 **exoALMA. XII. Weighing and Sizing exoALMA Disks with Rotation Curve Modelling**
Longarini, C., G. Lodato, G. Rosotti, et al.
The Astrophysical Journal, 984, L17

141 **exoALMA. V. Gaseous Emission Surfaces and Temperature Structures**
Galloway-Sprietsma, M., J. Bae, A. F. Izquierdo, et al.
The Astrophysical Journal, 984, L10

140 **exoALMA. VIII. Probabilistic Moment Maps and Data Products Using Nonparametric Linear Models**
Hilder, T., A. R. Casey, D. J. Price, et al.
The Astrophysical Journal, 984, L13

139 **exoALMA. I. Science Goals, Project Design, and Data Products**
Teague, R., M. Benisty, S. Facchini, et al.
The Astrophysical Journal, 984, L6

138 **exoALMA. II. Data Calibration and Imaging Pipeline**
Loomis, R. A., S. Facchini, M. Benisty, et al.
The Astrophysical Journal, 984, L7

137 **exoALMA. III. Line-intensity Modeling and System Property Extraction from Protoplanetary Disks**
Izquierdo, A. F., J. Stadler, M. Galloway-Sprietsma, et al.
The Astrophysical Journal, 984, L8

136 **exoALMA. XI. ALMA Observations and Hydrodynamic Models of LkCa 15: Implications for Planetary Mass Companions in the Dust Continuum Cavity**
Gardner, C. H., A. Isella, H. Li, et al.
The Astrophysical Journal, 984, L16

135 **exoALMA. XIII. Gas Masses from N_2H^+ and $C^{18}O$: A Comparison of Measurement Techniques for Protoplanetary Gas Disk Masses**
Trapman, L., C. Longarini, G. P. Rosotti, et al.
The Astrophysical Journal, 984, L18

134 **exoALMA. VI. Rotating under Pressure: Rotation Curves, Azimuthal Velocity Substructures, and Gas Pressure Variations**
Stadler, J., M. Benisty, A. J. Winter, et al.
The Astrophysical Journal, 984, L11

133 **exoALMA. VII. Benchmarking Hydrodynamics and Radiative Transfer Codes**
Bae, J., M. Flock, A. Izquierdo, et al.
The Astrophysical Journal, 984, L12

132 **exoALMA. IX. Regularized Maximum Likelihood Imaging of Non-Keplerian Features**
Zawadzki, B., I. Czekala, M. Galloway-Sprietsma, et al.
The Astrophysical Journal, 984, L14

131 **exoALMA. X. Channel Maps Reveal Complex ^{12}CO Abundance Distributions and a Variety of Kinematic Structures with Evidence for Embedded Planets**
Pinte, C., J. D. Ilee, J. Huang, et al.
The Astrophysical Journal, 984, L15

130 **exoALMA. IV. Substructures, Asymmetries, and the Faint Outer Disk in Continuum Emission**
Curone, P., S. Facchini, S. M. Andrews, et al.
The Astrophysical Journal, 984, L9

129 **Mapping the Merging Zone of Late Infall in the AB Aur Planet-forming System**
Speedie, J., R. Dong, **R. Teague**, et al.
The Astrophysical Journal, 981, L30

2024

14 PAPERS

128 **Chemistry in the GG Tau A Disk: Constraints from H_2D^+, N_2H^+, and DCO^+ High Angular Resolution ALMA Observations**
Kashyap, P., L. Majumdar, A. Dutrey, et al.
The Astrophysical Journal, 976, 258

127 **Kinematical signatures: Distinguishing between warps and radial flows**
Zuleta, A., T. Birnstiel, & **R. Teague**
Astronomy and Astrophysics, 692, A56

126 **Observational signatures of circumbinary discs - II. Kinematic signatures in velocity residuals**
Calcino, J., B. J. Norfolk, D. J. Price, et al.
Monthly Notices of the Royal Astronomical Society, 534, 2904

125 **Detection of Dimethyl Ether in the Central Region of the MWC 480 Protoplanetary Disk**
Yamato, Y., Y. Aikawa, V. V. Guzmán, et al.
The Astrophysical Journal, 974, 83

124 **Gravitational instability in a planet-forming disk**
Speedie, J., R. Dong, C. Hall, et al.
Nature, 633, 58

123 **ALMA high-resolution observations unveil planet formation shaping molecular emission in the PDS 70 disk**
Rampinelli, L., S. Facchini, M. Leemker, et al.
Astronomy and Astrophysics, 689, A65

122 **Outflow Driven by a Protoplanet Embedded in the TW Hya Disk**
Yoshida, T. C., H. Nomura, C. J. Law, et al.
The Astrophysical Journal, 971, L15

121 **On Kinematic Measurements of Self-gravity in Protoplanetary Disks**
Andrews, S. M., **R. Teague**, C. P. Wirth, et al.
The Astrophysical Journal, 970, 153

120 **Constraining the stellar masses and origin of the protostellar VLA 1623 system**
Sadavoy, S. I., P. Sheehan, J. J. Tobin, et al.
Astronomy and Astrophysics, 687, A308

119 **The First Spatially Resolved Detection of ^{13}CN in a Protoplanetary Disk and Evidence for Complex Carbon Isotope Fractionation**
Yoshida, T. C., H. Nomura, K. Furuya, et al.
The Astrophysical Journal, 966, 63

118 **Mapping the Vertical Gas Structure of the Planet-hosting PDS 70 Disk**
Law, C. J., M. Benisty, S. Facchini, et al.
The Astrophysical Journal, 964, 190

117 **The Carbon Isotopic Ratio and Planet Formation**
Bergin, E. A., A. Bosman, **R. Teague**, et al.
The Astrophysical Journal, 965, 147

116 **High turbulence in the IM Lup protoplanetary disk. Direct observational constraints from CN and C_2H emission**
Paneque-Carreño, T., A. F. Izquierdo, **R. Teague**, et al.
Astronomy and Astrophysics, 684, A174

115 **JWST-MIRI Spectroscopy of Warm Molecular Emission and Variability in the AS 209 Disk**
Romero-Mirza, C. E., K. I. Öberg, A. Banzatti, et al.
The Astrophysical Journal, 964, 36

2023
17 PAPERS

114 **Protoplanetary disks in K_s-band total intensity and polarized light**
Ren, B. B., M. Benisty, C. Ginski, et al.
Astronomy and Astrophysics, 680, A114

113 **Three-dimensional magnetic field imaging of protoplanetary disks using Zeeman broadening and linear polarization observations**
Lankhaar, B. & **R. Teague**
Astronomy and Astrophysics, 678, A17

112 **MagAO-X and HST High-contrast Imaging of the AS209 Disk at Hα**
Cugno, G., Y. Zhou, T. Thanathibodee, et al.
The Astronomical Journal, 166, 162

111 **MAPS: Constraining Serendipitous Time Variability in Protoplanetary Disk Molecular Ion Emission**
Waggoner, A. R., L. I. Cleeves, R. A. Loomis, et al.
The Astrophysical Journal, 956, 103

110 **A Magnetically Driven Disk Wind in the Inner Disk of PDS 70**
Campbell-White, J., C. F. Manara, M. Benisty, et al.
The Astrophysical Journal, 956, 25

109 **Constraining the gas distribution in the PDS 70 disc as a method to assess the effect of planet-disc interactions**
Portilla-Revelo, B., I. Kamp, S. Facchini, et al.
Astronomy and Astrophysics, 677, A76

108 **Implications for Chondrule Formation Regions and Solar Nebula Magnetism from Statistical Reanalysis of Chondrule Paleomagnetism**
Fu, R. R., S. C. Steele, J. B. Simon, et al.
The Planetary Science Journal, 4, 151

107 **Observational signatures of circumbinary discs - I. Kinematics**
Calcino, J., D. J. Price, C. Pinte, et al.
Monthly Notices of the Royal Astronomical Society, 523, 5763

106 **Tentative co-orbital submillimeter emission within the Lagrangian region L_5 of the protoplanet PDS 70 b**
Balsalobre-Ruza, O., I. de Gregorio-Monsalvo, J. Lillo-Box, et al.
Astronomy and Astrophysics, 675, A172

105 **Kinematic Structures in Planet-Forming Disks**
Pinte, C., **R. Teague**, K. Flaherty, et al.
Protostars and Planets VII, 534, 645

104 **Molecules with ALMA at Planet-forming Scales (MAPS): Complex Kinematics in the AS 209 Disk Induced by a Forming Planet and Disk Winds**
Galloway-Sprietsma, M., J. Bae, **R. Teague**, et al.
The Astrophysical Journal, 950, 147

103 **Mapping Protoplanetary Disk Vertical Structure with CO Isotopologue Line Emission**
Law, C. J., **R. Teague**, K. I. Öberg, et al.
The Astrophysical Journal, 948, 60

102 **An infrared transient from a star engulfing a planet**
De, K., M. MacLeod, V. Karambelkar, et al.
Nature, 617, 55

101 **A kinematically detected planet candidate in a transition disk**
Stadler, J., M. Benisty, A. Izquierdo, et al.
Astronomy and Astrophysics, 670, L1

100 **Kinematics and brightness temperatures of transition discs. A survey of gas substructures as seen with ALMA**
Wölfer, L., S. Facchini, N. van der Marel, et al.
Astronomy and Astrophysics, 670, A154

99 **Cold Deuterium Fractionation in the Nearest Planet-forming Disk**
Romero-Mirza, C. E., K. I. Öberg, C. J. Law, et al.
The Astrophysical Journal, 943, 35

98 **UV-driven chemistry as a signpost of late-stage planet formation**
Calahan, J. K., E. A. Bergin, A. D. Bosman, et al.
Nature Astronomy, 7, 49

2022

10 PAPERS

97 **A Localized Kinematic Structure Detected in Atomic Carbon Emission Spatially Coincident with a Proposed Protoplanet in the HD 163296 Disk**
Alarcón, F., E. A. Bergin, & **R. Teague**
The Astrophysical Journal, 941, L24

96 **A kinematic excess in the annular gap and gas-depleted cavity in the disc around HD 169142**
Garg, H., C. Pinte, I. Hammond, et al.
Monthly Notices of the Royal Astronomical Society, 517, 5942

95 **Unveiling the outer dust disc of TW Hya with deep ALMA observations**
Ilee, J. D., C. Walsh, J. Jennings, et al.
Monthly Notices of the Royal Astronomical Society, 515, L23

94 **Mapping the Complex Kinematic Substructure in the TW Hya Disk**
Teague, R., J. Bae, S. M. Andrews, et al.
The Astrophysical Journal, 936, 163

93 **Molecules with ALMA at Planet-forming Scales (MAPS): A Circumplanetary Disk Candidate in Molecular-line Emission in the AS 209 Disk**
Bae, J., **R. Teague**, S. M. Andrews, et al.
The Astrophysical Journal, 934, L20

92 **CO Line Emission Surfaces and Vertical Structure in Midinclination Protoplanetary Disks**
Law, C. J., S. Crystian, **R. Teague**, et al.
The Astrophysical Journal, 932, 114

91 **Gas and Dust Shadows in the TW Hydrae Disk**
Teague, R., J. Bae, M. Benisty, et al.
The Astrophysical Journal, 930, 144

90 **Polarization from Aligned Dust Grains in the β Pic Debris Disk**
Hull, C. L. H., H. Yang, P. C. Cortés, et al.
The Astrophysical Journal, 930, 49

89 **Gas Disk Sizes from CO Line Observations: A Test of Angular Momentum Evolution**
Long, F., S. M. Andrews, G. Rosotti, et al.
The Astrophysical Journal, 931, 6

88 **Probing inner and outer disk misalignments in transition disks. Constraints from VLTI/GRAVITY and ALMA observations**
Bohn, A. J., M. Benisty, K. Perraut, et al.
Astronomy and Astrophysics, 658, A183

2021
40 PAPERS

87 **Discovery of Molecular-line Polarization in the Disk of TW Hya**
Teague, R., C. L. H. Hull, S. Guilloteau, et al.
The Astrophysical Journal, 922, 139

86 **disksurf: Extracting the 3D Structure of Protoplanetary Disks**
Teague, R., C. Law, J. Huang, et al.
The Journal of Open Source Software, 6, 3827

85 **The First Detection of CH_2CN in a Protoplanetary Disk**
Canta, A., **R. Teague**, R. Le Gal, et al.
The Astrophysical Journal, 922, 62

84 **Molecules with ALMA at Planet-forming Scales (MAPS). X. Studying Deuteration at High Angular Resolution toward Protoplanetary Disks**
Cataldi, G., Y. Yamato, Y. Aikawa, et al.
The Astrophysical Journal Supplement Series, 257, 10

83 **Molecules with ALMA at Planet-forming Scales (MAPS). XI. CN and HCN as Tracers of Photochemistry in Disks**
Bergner, J. B., K. I. Öberg, V. V. Guzmán, et al.
The Astrophysical Journal Supplement Series, 257, 11

82 **Molecules with ALMA at Planet-forming Scales (MAPS). XIII. HCO^+ and Disk Ionization Structure**
Aikawa, Y., G. Cataldi, Y. Yamato, et al.
The Astrophysical Journal Supplement Series, 257, 13

81 **Molecules with ALMA at Planet-forming Scales (MAPS). XVI. Characterizing the Impact of the Molecular Wind on the Evolution of the HD 163296 System**
Booth, A. S., B. Tabone, J. D. Ilee, et al.
The Astrophysical Journal Supplement Series, 257, 16

80 **Molecules with ALMA at Planet-forming Scales (MAPS). XIX. Spiral Arms, a Tail, and Diffuse Structures Traced by CO around the GM Aur Disk**
Huang, J., E. A. Bergin, K. I. Öberg, et al.
The Astrophysical Journal Supplement Series, 257, 19

79 **Molecules with ALMA at Planet-forming Scales. XX. The Massive Disk around GM Aurigae**
Schwarz, K. R., J. K. Calahan, K. Zhang, et al.
The Astrophysical Journal Supplement Series, 257, 20

78 **Molecules with ALMA at Planet-forming Scales (MAPS). VI. Distribution of the Small Organics HCN, C_2H, and H_2CO**
Guzmán, V. V., J. B. Bergner, C. J. Law, et al.
The Astrophysical Journal Supplement Series, 257, 6

77 **Molecules with ALMA at Planet-forming Scales (MAPS). VIII. CO Gap in AS 209-Gas Depletion or Chemical Processing?**
Alarcón, F., A. D. Bosman, E. A. Bergin, et al.
The Astrophysical Journal Supplement Series, 257, 8

76 **Molecules with ALMA at Planet-forming Scales (MAPS). XVII. Determining the 2D Thermal Structure of the HD 163296 Disk**
Calahan, J. K., E. A. Bergin, K. Zhang, et al.
The Astrophysical Journal Supplement Series, 257, 17

75 **Molecules with ALMA at Planet-forming Scales (MAPS). III. Characteristics of Radial Chemical Substructures**
Law, C. J., R. A. Loomis, **R. Teague**, et al.
The Astrophysical Journal Supplement Series, 257, 3

74 **Molecules with ALMA at Planet-forming Scales (MAPS). IV. Emission Surfaces and Vertical Distribution of Molecules**
Law, C. J., **R. Teague**, R. A. Loomis, et al.
The Astrophysical Journal Supplement Series, 257, 4

73 **Molecules with ALMA at Planet-forming Scales (MAPS). XVIII. Kinematic Substructures in the Disks of HD 163296 and MWC 480**
Teague, R., J. Bae, Y. Aikawa, et al.
The Astrophysical Journal Supplement Series, 257, 18

72 **Molecules with ALMA at Planet-forming Scales (MAPS). XIV. Revealing Disk Substructures in Multiwavelength Continuum Emission**
Sierra, A., L. M. Pérez, K. Zhang, et al.
The Astrophysical Journal Supplement Series, 257, 14

71 **Molecules with ALMA at Planet-forming Scales (MAPS). XV. Tracing Protoplanetary Disk Structure within 20 au**
Bosman, A. D., E. A. Bergin, R. A. Loomis, et al.
The Astrophysical Journal Supplement Series, 257, 15

70 **Molecules with ALMA at Planet-forming Scales (MAPS). I. Program Overview and Highlights**
Öberg, K. I., V. V. Guzmán, C. Walsh, et al.
The Astrophysical Journal Supplement Series, 257, 1

69 **Molecules with ALMA at Planet-forming Scales (MAPS). II. CLEAN Strategies for Synthesizing Images of Molecular Line Emission in Protoplanetary Disks**
Czekala, I., R. A. Loomis, **R. Teague**, et al.
The Astrophysical Journal Supplement Series, 257, 2

68 **Molecules with ALMA at Planet-forming Scales (MAPS). V. CO Gas Distributions**
Zhang, K., A. S. Booth, C. J. Law, et al.
The Astrophysical Journal Supplement Series, 257, 5

67 **Molecules with ALMA at Planet-forming Scales (MAPS). VII. Substellar O/H and C/H and Superstellar C/O in Planet-feeding Gas**
Bosman, A. D., F. Alarcón, E. A. Bergin, et al.
The Astrophysical Journal Supplement Series, 257, 7

66 **Molecules with ALMA at Planet-forming Scales (MAPS). IX. Distribution and Properties of the Large Organic Molecules HC_3N, CH_3CN, and c-C_3H_2**
Ilee, J. D., C. Walsh, A. S. Booth, et al.
The Astrophysical Journal Supplement Series, 257, 9

65 **Molecules with ALMA at Planet-forming Scales (MAPS). XII. Inferring the C/O and S/H Ratios in Protoplanetary Disks with Sulfur Molecules**
Le Gal, R., K. I. Öberg, **R. Teague**, et al.
The Astrophysical Journal Supplement Series, 257, 12

64 **Mapping the 3D Kinematical Structure of the Gas Disk of HD 169142**
Yu, H., **R. Teague**, J. Bae, et al.
The Astrophysical Journal, 920, L33

63 **The Chemical Inventory of the Planet-hosting Disk PDS 70**
Facchini, S., **R. Teague**, J. Bae, et al.
The Astronomical Journal, 162, 99

62 **Limits on Millimeter Continuum Emission from Circumplanetary Material in the DSHARP Disks**
Andrews, S. M., W. Elder, S. Zhang, et al.
The Astrophysical Journal, 916, 51

61 **The Architecture of the V892 Tau System: The Binary and Its Circumbinary Disk**
Long, F., S. M. Andrews, J. Vega, et al.
The Astrophysical Journal, 915, 131

60 **A Circumplanetary Disk around PDS70c**
Benisty, M., J. Bae, S. Facchini, et al.
The Astrophysical Journal, 916, L2

59 **Investigating the Relative Gas and Small Dust Grain Surface Heights in Protoplanetary Disks**
Rich, E. A., **R. Teague**, J. D. Monnier, et al.
The Astrophysical Journal, 913, 138

58 **Vortex-like kinematic signal, spirals, and beam smearing effect in the HD 142527 disk**
Boehler, Y., F. Ménard, C. M. T. Robert, et al.
Astronomy and Astrophysics, 650, A59

57 **Observational Signature of Tightly Wound Spirals Driven by Buoyancy Resonances in Protoplanetary Disks**
Bae, J., **R. Teague**, & Z. Zhu
The Astrophysical Journal, 912, 56

56 **A highly non-Keplerian protoplanetary disc. Spiral structure in the gas disc of CQ Tau**
Wölfer, L., S. Facchini, N. T. Kurtovic, et al.
Astronomy and Astrophysics, 648, A19

55 **An Atacama Large Millimeter/submillimeter Array Survey of Chemistry in Disks around M4-M5 Stars**
Pegues, J., K. I. Öberg, J. B. Bergner, et al.
The Astrophysical Journal, 911, 150

54 **The TW Hya Rosetta Stone Project IV: A Hydrocarbon-rich Disk Atmosphere**
Cleeves, L. I., R. A. Loomis, **R. Teague**, et al.
The Astrophysical Journal, 911, 29

53 **ALMA CN Zeeman Observations of AS 209: Limits on Magnetic Field Strength and Magnetically Driven Accretion Rate**
Harrison, R. E., L. W. Looney, I. W. Stephens, et al.
The Astrophysical Journal, 908, 141

52 **The TW Hya Rosetta Stone Project. III. Resolving the Gaseous Thermal Profile of the Disk**
Calahan, J. K., E. Bergin, K. Zhang, et al.
The Astrophysical Journal, 908, 8

51 **Dynamical Masses and Stellar Evolutionary Model Predictions of M Stars**
Pegues, J., I. Czekala, S. M. Andrews, et al.
The Astrophysical Journal, 908, 42

50 **The TW Hya Rosetta Stone Project. I. Radial and Vertical Distributions of DCN and DCO^+**

Öberg, K. I., L. I. Cleeves, J. B. Bergner, et al.
The Astronomical Journal, 161, 38

49 **The TW Hya Rosetta Stone Project. II. Spatially Resolved Emission of Formaldehyde Hints at Low-temperature Gas-phase Formation**

Terwisscha van Scheltinga, J., M. R. Hogerheijde, L. I. Cleeves, et al.
The Astrophysical Journal, 906, 111

48 **ALMA chemical survey of disk-outflow sources in Taurus (ALMA-DOT). V. Sample, overview, and demography of disk molecular emission**

Garufi, A., L. Podio, C. Codella, et al.
Astronomy and Astrophysics, 645, A145

2020

17 PAPERS

47 **ALMA chemical survey of disk-outflow sources in Taurus (ALMA-DOT). III. The interplay between gas and dust in the protoplanetary disk of DG Tau**

Podio, L., A. Garufi, C. Codella, et al.
Astronomy and Astrophysics, 644, A119

46 **Chemical Evolution in a Protoplanetary Disk within Planet Carved Gaps and Dust Rings**

Alarcón, F., **R. Teague**, K. Zhang, et al.
The Astrophysical Journal, 905, 68

45 **Predicting the Kinematic Evidence of Gravitational Instability**

Hall, C., R. Dong, **R. Teague**, et al.
The Astrophysical Journal, 904, 148

44 **GoFish: Molecular line detections in protoplanetary disks**

Teague, R.
Astrophysics Source Code Library, ascl:2011.016

43 **ALMA and VLA Observations of EX Lupi in Its Quiescent State**

White, J. A., Á. Kóspál, A. G. Hughes, et al.
The Astrophysical Journal, 904, 37

42 **Visualizing the Kinematics of Planet Formation**

Disk Dynamics Collaboration, P. J. Armitage, J. Bae, et al.
arXiv e-prints, arXiv:2009.04345

41 **Low-level Carbon Monoxide Line Polarization in Two Protoplanetary Disks: HD 142527 and IM Lup**

Stephens, I. W., M. Fernández-López, Z.-Y. Li, et al.
The Astrophysical Journal, 901, 71

40 **The Excitation Conditions of CN in TW Hya**
Teague, R. & R. Loomis
The Astrophysical Journal, 899, 157

39 **Annular substructures in the transition disks around LkCa 15 and J1610**
Facchini, S., M. Benisty, J. Bae, et al.
Astronomy and Astrophysics, 639, A121

38 **A three-dimensional view of Gomez's hamburger**
Teague, R., M. R. Jankovic, T. J. Haworth, et al.
Monthly Notices of the Royal Astronomical Society, 495, 451

37 **Hints of a Population of Solar System Analog Planets from ALMA**
Long, D. E., K. Zhang, **R. Teague**, et al.
The Astrophysical Journal, 895, L46

36 **The efficiency of dust trapping in ringed protoplanetary discs**
Rosotti, G. P., **R. Teague**, C. Dullemond, et al.
Monthly Notices of the Royal Astronomical Society, 495, 173

35 **ALMA chemical survey of disk-outflow sources in Taurus (ALMA-DOT). I. CO, CS, CN, and H_2CO around DG Tau B**
Garufi, A., L. Podio, C. Codella, et al.
Astronomy and Astrophysics, 636, A65

34 **A Multifrequency ALMA Characterization of Substructures in the GM Aur Protoplanetary Disk**
Huang, J., S. M. Andrews, C. P. Dullemond, et al.
The Astrophysical Journal, 891, 48

33 **Spiral arms in the protoplanetary disc HD100453 detected with ALMA: evidence for binary-disc interaction and a vertical temperature gradient**
Rosotti, G. P., M. Benisty, A. Juhász, et al.
Monthly Notices of the Royal Astronomical Society, 491, 1335

32 **Tracing the physical conditions of planet formation with molecular excitation**
Teague, R.
Laboratory Astrophysics: From Observations to Interpretation, 350, 181

31 **Accretion disks around young stars: the cradles of planet formation**
Semenov, D. A. & **R. D. Teague**
Europhysics News, 51, 29

2019
12 PAPERS

30 **Meridional flows in the disk around a young star**
Teague, R., J. Bae, & E. A. Bergin
Nature, 574, 378

29 **Spiral Structure in the Gas Disk of TW Hya**
Teague, R., J. Bae, J. Huang, et al.
The Astrophysical Journal, 884, L56

28 **An Ideal Testbed for Planet-Disk Interaction: Two Giant Protoplanets in Resonance Shaping the PDS 70 Protoplanetary Disk**
Bae, J., Z. Zhu, C. Baruteau, et al.
The Astrophysical Journal, 884, L41

27 **GoFish: Fishing for Line Observations in Protoplanetary Disks**
Teague, R.
The Journal of Open Source Software, 4, 1632

26 **ALMA observations of A0620-00: fresh clues on the nature of quiescent black hole X-ray binary jets**
Gallo, E., **R. Teague**, R. M. Plotkin, et al.
Monthly Notices of the Royal Astronomical Society, 488, 191

25 **Detection of Continuum Submillimeter Emission Associated with Candidate Protoplanets**
Isella, A., M. Benisty, **R. Teague**, et al.
The Astrophysical Journal, 879, L25

24 **Statistical Uncertainties in Moment Maps of Line Emission**
Teague, R.
Research Notes of the American Astronomical Society, 3, 74

23 **Realizing the Unique Potential of ALMA to Probe the Gas Reservoir of Planet Formation**
Cleeves, I., R. Loomis, **R. Teague**, et al.
Bulletin of the American Astronomical Society, 51, 81

22 **Planet formation — The case for large efforts on the computational side**
Lyra, W., T. Haworth, B. Bitsch, et al.
Bulletin of the American Astronomical Society, 51, 129

21 **Highly structured disk around the planet host PDS 70 revealed by high-angular resolution observations with ALMA**
Keppler, M., **R. Teague**, J. Bae, et al.
Astronomy and Astrophysics, 625, A118

20 **Line Ratios Reveal N_2H^+ Emission Originates above the Midplane in TW Hydrae**
Schwarz, K. R., **R. Teague**, & E. A. Bergin
The Astrophysical Journal, 876, L13

19 **eddy: Extracting Protoplanetary Disk Dynamics with Python**
Teague, R.
The Journal of Open Source Software, 4, 1220

2018

7 PAPERS

18 **Evidence for a Vertical Dependence on the Pressure Structure in AS 209**

Teague, R., J. Bae, T. Birnstiel, et al.

The Astrophysical Journal, 868, 113

17 **Temperature, Mass, and Turbulence: A Spatially Resolved Multiband Non-LTE Analysis of CS in TW Hya**

Teague, R., T. Henning, S. Guilloteau, et al.

The Astrophysical Journal, 864, 133

16 **Chemistry in disks. XI. Sulfur-bearing species as tracers of protoplanetary disk physics and chemistry: the DM Tau case**

Semenov, D., C. Favre, D. Fedele, et al.

Astronomy and Astrophysics, 617, A28

15 **A Robust Method to Measure Centroids of Spectral Lines**

Teague, R. & D. Foreman-Mackey

Research Notes of the American Astronomical Society, 2, 173

14 **A Kinematical Detection of Two Embedded Jupiter-mass Planets in HD 163296**

Teague, R., J. Bae, E. A. Bergin, et al.

The Astrophysical Journal, 860, L12

13 **Turbulence in the TW Hya Disk**

Flaherty, K. M., A. M. Hughes, **R. Teague**, et al.

The Astrophysical Journal, 856, 117

12 **ALMA continuum observations of the protoplanetary disk AS 209. Evidence of multiple gaps opened by a single planet**

Fedele, D., M. Tazzari, R. Booth, et al.

Astronomy and Astrophysics, 610, A24

2017

7 PAPERS

11 **Radiation Hydrodynamical Turbulence in Protoplanetary Disks: Numerical Models and Observational Constraints**

Flock, M., R. P. Nelson, N. J. Turner, et al.

The Astrophysical Journal, 850, 131

10 **The Flying Saucer: Tomography of the thermal and density gas structure of an edge-on protoplanetary disk**

Dutrey, A., S. Guilloteau, V. Piétu, et al.

Astronomy and Astrophysics, 607, A130

9 **Multiplicity and disks within the high-mass core NGC 7538IRS1. Resolving cm line and continuum emission at 0.06″ × 0.05″ resolution**

Beuther, H., H. Linz, T. Henning, et al.

Astronomy and Astrophysics, 605, A61

8 **On the methanol emission detection in the TW Hya disc: the role of grain surface chemistry and non-LTE excitation**
Parfenov, S. Y., D. A. Semenov, T. Henning, et al.
Monthly Notices of the Royal Astronomical Society, 468, 2024

7 **Three Radial Gaps in the Disk of TW Hydrae Imaged with SPHERE**
van Boekel, R., T. Henning, J. Menu, et al.
The Astrophysical Journal, 837, 132

6 **A Surface Density Perturbation in the TW Hydrae Disk at 95 au Traced by Molecular Emission**
Teague, R., D. Semenov, U. Gorti, et al.
The Astrophysical Journal, 835, 228

5 **Tracing the earliest stages of planet formation through modelling and sub-mm observations**
Teague, R.
Ph.D. Thesis

2016
3 PAPERS

4 **Grand Challenges in Protoplanetary Disc Modelling**
Haworth, T. J., J. D. Ilee, D. H. Forgan, et al.
Publications of the Astronomical Society of Australia, 33, e053

3 **Inferring the evolutionary stages of the internal structures of NGC 7538 S and IRS1 from chemistry**
Feng, S., H. Beuther, D. Semenov, et al.
Astronomy and Astrophysics, 593, A46

2 **Measuring turbulence in TW Hydrae with ALMA: methods and limitations**
Teague, R., S. Guilloteau, D. Semenov, et al.
Astronomy and Astrophysics, 592, A49

2015
1 PAPERS

1 **Chemistry in disks. IX. Observations and modelling of HCO^+ and DCO^+ in DM Tauri**
Teague, R., D. Semenov, S. Guilloteau, et al.
Astronomy and Astrophysics, 574, A137